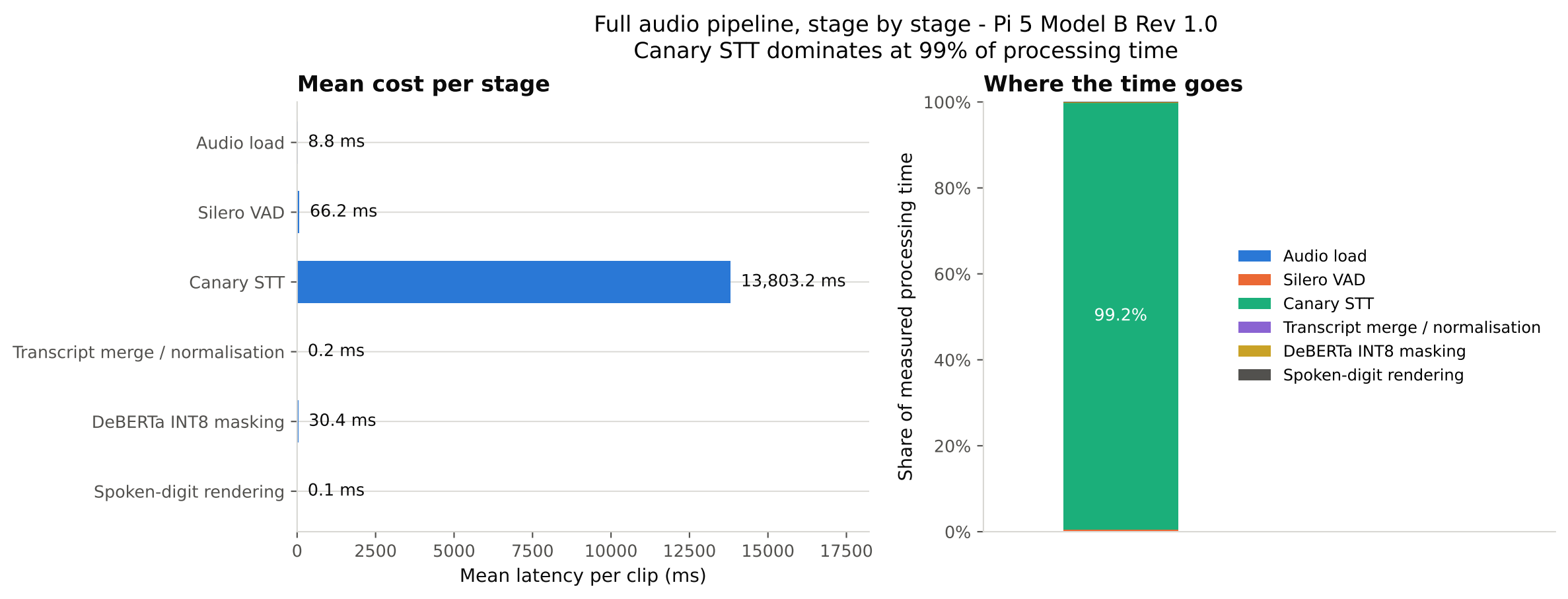
Full audio pipeline, stage by stage - Pi 5 Model B Rev 1.0
Canary STT dominates at 99% of processing time
Mean cost per stage
Audio load
Silero VAD
Canary STT
Transcript merge / normalisation
DeBERTa INT8 masking
Spoken-digit rendering
8.8 ms
66.2 ms
13,803.2 ms
0.2 ms
30.4 ms
0.1 ms
0
2500
5000
7500
10000
12500
15000
17500
Mean latency per clip (ms)
Where the time goes
Share of measured processing time
100%
80%
60%
40%
20%
0%
99.2%
Audio load
Silero VAD
Canary STT
Transcript merge / normalisation
DeBERTa INT8 masking
Spoken-digit rendering